

2. Requirement Analysis Phase

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

- **Functional Requirements:** User input collection; AI-based workout-plan generation; nutrition and recovery suggestions; feedback-based plan modification; storage of user information and generated plans; administrator access to stored information.
- **Non-Functional Requirements:** Usable interface, reliable API processing, maintainable code, secure handling of configuration values, reasonable response time, and clear error handling.
- **Software Requirements:** Python, FastAPI, Google Gemini API, Jinja2, HTML, CSS, SQLite, SQLAlchemy, Uvicorn, and a code editor such as Visual Studio Code.
- **Hardware Requirements:** A computer with a modern web browser, internet connectivity for AI API access, and sufficient storage for the application and database.
- **Inputs and Outputs:** Inputs include fitness-related user information, goals, intensity, and feedback. Outputs include a personalized 7-day plan, nutrition/recovery guidance, and updated plans after feedback.